A REPRESENTABLE FUNCTOR IS AN ALGEBRAIC SPACE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is short but is a complete verification of the three defining conditions of an algebraic space, with each step stated; nothing is omitted and no single citation carries the claim.

Editorial note. The single reference of the counted spans is recorded as a non-load-bearing navigation pointer; the argument that consumes it stays inside the counted proof span.

Editorial note. Preamble: sources/stacks-spaces/retained-preamble.tex, sliced mechanically from the chapter's own preamble.tex.

Allowed standard prerequisites (editorial).

- Representable presheaves on the fppf site are sheaves. remarks of Section section-general, used at proof lines 1074-1075. Supplies the first defining condition of an algebraic space.
- The fppf site has fibre products and products of pairs. used at proof lines 1076-1077. Makes the diagonal h_T -> h_T x h_T = h_{T x T} representable.
- Definition of an algebraic space. Definition definition-algebraic-space, used at proof lines 1074-1078. Fixes the three conditions that are checked.

Source reference. Original section 025T: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2826–2826. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *A scheme is an algebraic space. More precisely, given a scheme $T \in \mathrm{Ob}((Sch/S)_{fppf})$ the representable functor h_T is an algebraic space.*

Proof. The functor h_T is a sheaf by our remarks in Section 025T. The diagonal $h_T \to h_T \times h_T = h_{T \times T}$ is representable because $(Sch/S)_{fppf}$ has fibre products. The identity map $h_T \to h_T$ is surjective étale. □

Date: stacks-project:spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.